CSOC WG

AWS Cost Savings

Intro

Focus: Reducing Gen3 operating costs.

Goal: Run multiple secure, FedRAMP-compliant AWS environments for under $1,000/month.

Status: Platform Engineering is actively iterating and making steady progress toward this target.

Agenda

Karpenter
Compute provisioning

Cilium
Overlay networking

Shrinking Portal
Build once - serve many!

AWS Cost Usage Reports
Quick tips on how to use them?

Karpenter + Spot Instances for Cost Savings

Goal: Reduce AWS compute costs (~70%) for Kubernetes workloads

Technologies:

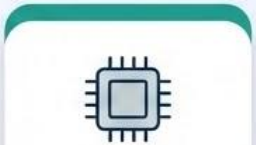
- **Karpenter:** Dynamically provisions right-sized nodes
- **Spot Instances:** Low-cost, interruptible compute

Strategy:

- Use Spot capacity
- Enable flexible instance categories (c / m / r)
- Apply consolidation policies
- Ensure uptime with Replicas & PDBs

Instance Categories (Flexibility = Savings)

- Enabled multiple instance families:



c (compute optimized)

m (general purpose)

r (memory optimized)

👉 **Result:**

- Higher chance of getting Spot capacity
- Lower cost via flexible scheduling

Avoided locking into specific instance types

Enforcing Spot Usage (Karpenter)

Configuration

```
- key: karpenter.sh/capacity-type
  operator: In
  values:
    - spot
```

👉 **Benefits:**

- Ensures workloads only run on Spot nodes
- Prevents accidental use of on-demand capacity

Instance Sizing Constraints

```
- key: karpenter.k8s.aws/instance-memory
  operator: Gt
  values:
    - '7000'
- key: karpenter.k8s.aws/instance-memory
  operator: Lt
  values:
    - '32500'
```

- Restricts nodes to ~7GB – 32GB range
- Avoids:
  - Very small inefficient nodes
  - Very large expensive nodes

👍 Better cost vs utilization balance

Consolidation (Major Cost Saver)

- **Removes:**
  - Empty nodes
  - Underutilized nodes
- **Replaces nodes with cheaper alternatives**

👉 **Aggressive optimization with controlled disruption**

Handling Spot Interruptions (Uptime)

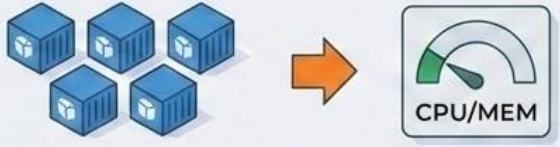
- Spot instances **can be terminated anytime**.

What we could implement:

- Replicas for all workloads
- Pod Disruption Budgets (PDBs)
- Workload separation:
  - Critical → On-demand
  - Non-critical → Spot

👉 **Ensures availability despite interruptions**

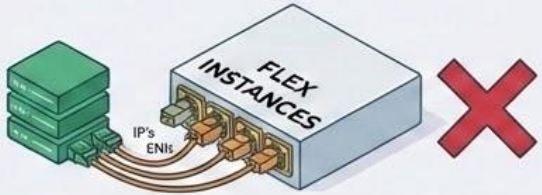
Cilium – The problem



PODS OPTIMIZED: Reduced CPU/Memory requests.

POD DENSITY LIMIT REACHED

CAUSE: EC2 instance limits on ENI's and real IP's.

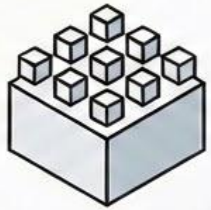


STILL LIMITED BY POD DENSITY: Flex instances not enough.

Cilium - The solution

cilium

Cilium uses Virtual IP addresses for pod networking, not host IPs.



Enables a high Pod Limit of 2000/node, which is very unlikely to be reached.

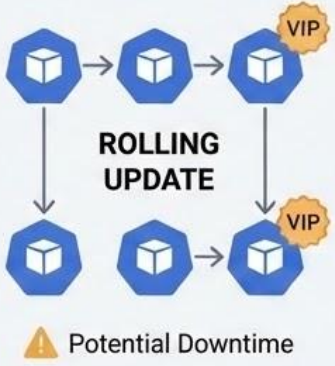
This allows for Tighter Node Packing, running more pods per node efficiently.

Cilium - How to setup

1

Enable in cluster-level resources chart.

2

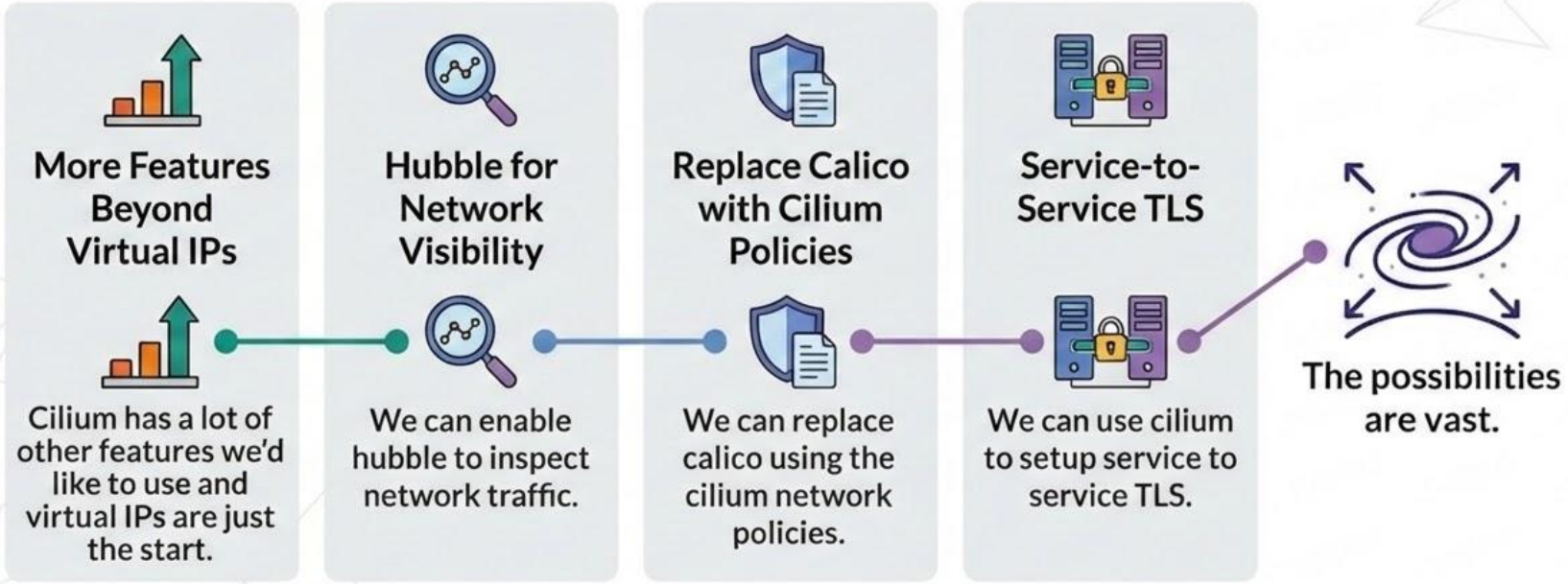


Roll pods to use Virtual IPs.

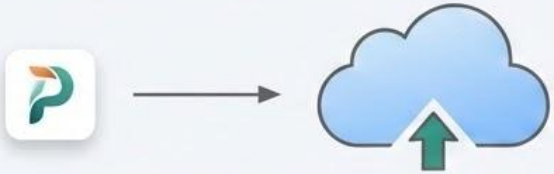
3

Optionally remove old daemonsets; utilize Cilium.

Cilium - Next steps



Shrinking Portal - The problem



High Memory at Startup (Build)

Portal is technically a **very lightweight** app that requires a lot of memory at startup to build the application.

We need this because we need to build very different frontends based on any differences in the configuration files used, or even portal version.

Faster Builds = More CPU: We also want to have portal build slightly faster, so we end up giving more CPU which is only really needed at build time.

Failed Docker Image Approach: We tried to build individual docker images based on specific builds, but it became hard to manage and was not widely adopted.

Large Memory Requirement:
There is a need for a large amount of memory at build time, which ends up having us require a lot of memory.

Shrinking Portal - The solution

Cache Build Objects

Instead of building each time, cache build objects. Create new builds only if changes found.

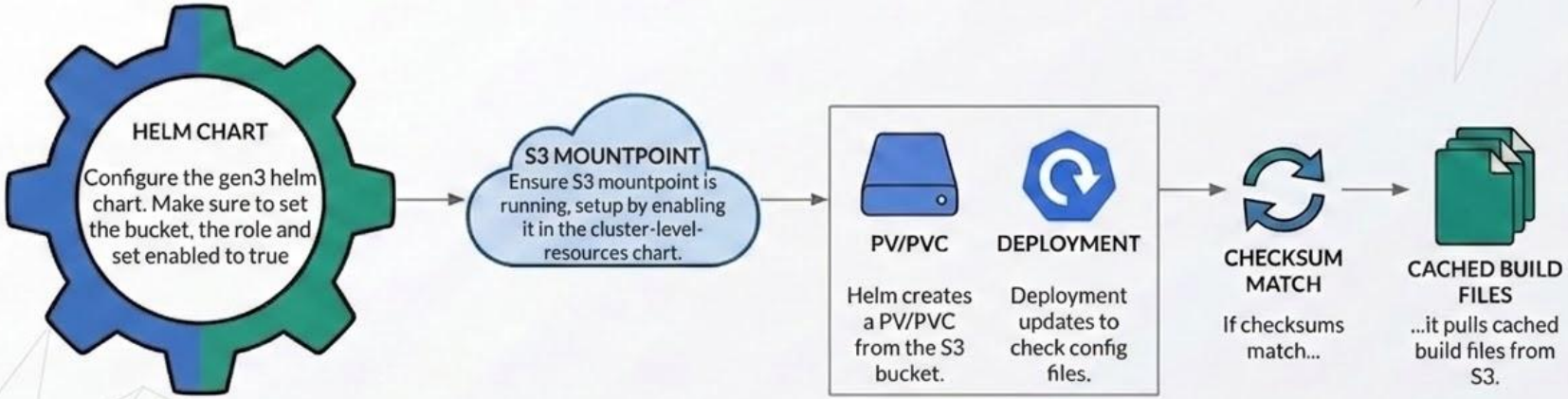
Verify with Checksums

Utilize checksums of configuration to verify cached config. Pull from cache based on checksum.

S3 Mountpoint Backend

Build cache using mountpoint as backend. Allows multiple pods to use same PVC, not AZ-locked. Requires S3 mountpoint & portal role with bucket access.

Shrinking Portal - How to use it



Rethinking Stateful Services

Kubernetes vs. Managed Cloud Services

Core Question: Do our metrics and resource demands justify the premium price of managed services? *(Especially in dev/test/staging environments, lower tier prod environments)*

Strategic Shift: Re-evaluating reliance on fully managed stateful services to drive down operational costs in lower tier environments.

Key Focus Areas: Elasticsearch and PostgreSQL.

Running PostgreSQL in Kubernetes

Single Pod:

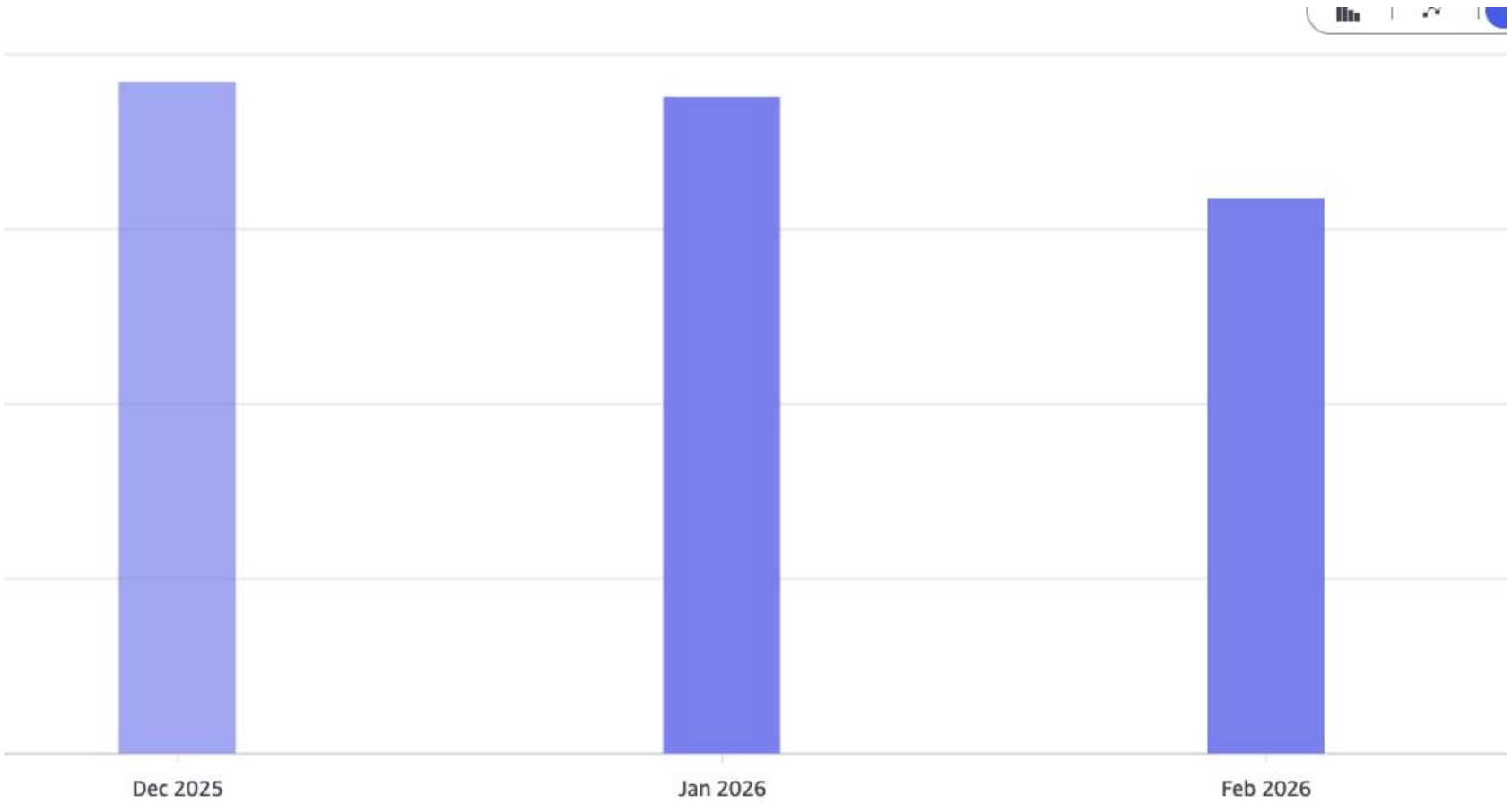
- Bitnami Charts
- Easy to spin up.
- Good for dev/test - used in our CI!
- Out of support

Operator

- CloudnativePG
- https://cloudnative-pg.io/

AWS Cost / usage reports

- How to analyze costs
- Find and clean up stale resources
    - Practice good cloud hygiene!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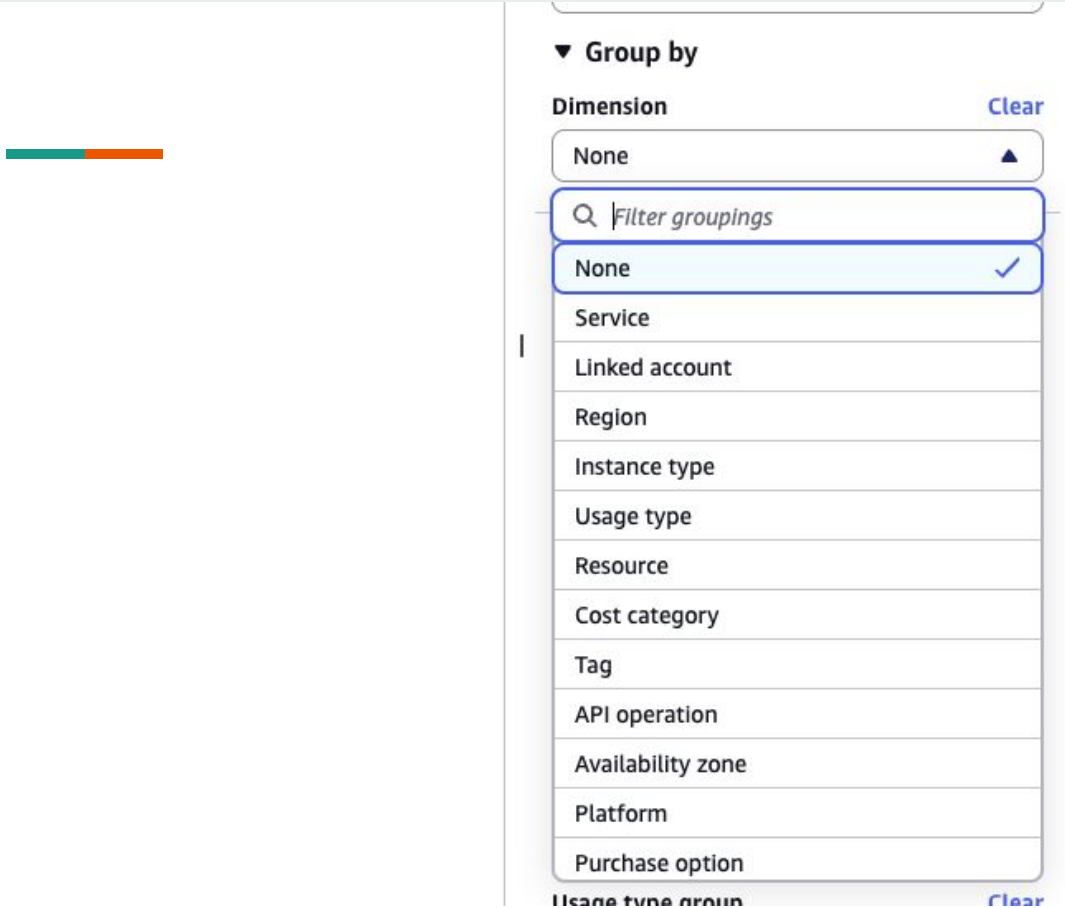
Jan 2026
Feb 2026


Group by
Dimension
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Service
Linked account
Region
Instance type
Usage type
Resource
Cost category
Tag
API operation
Availability zone
Platform
Purchase option
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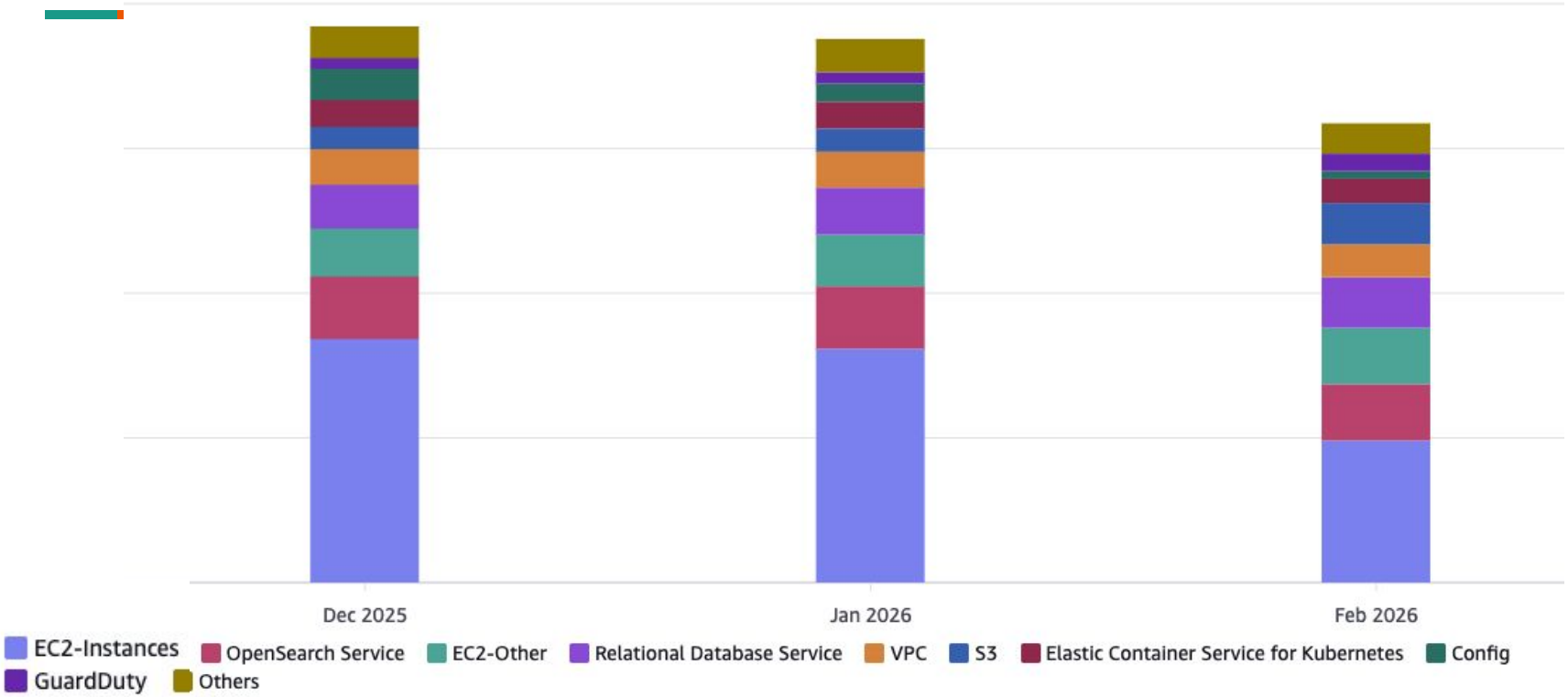
Dec 2025
Jan 2026
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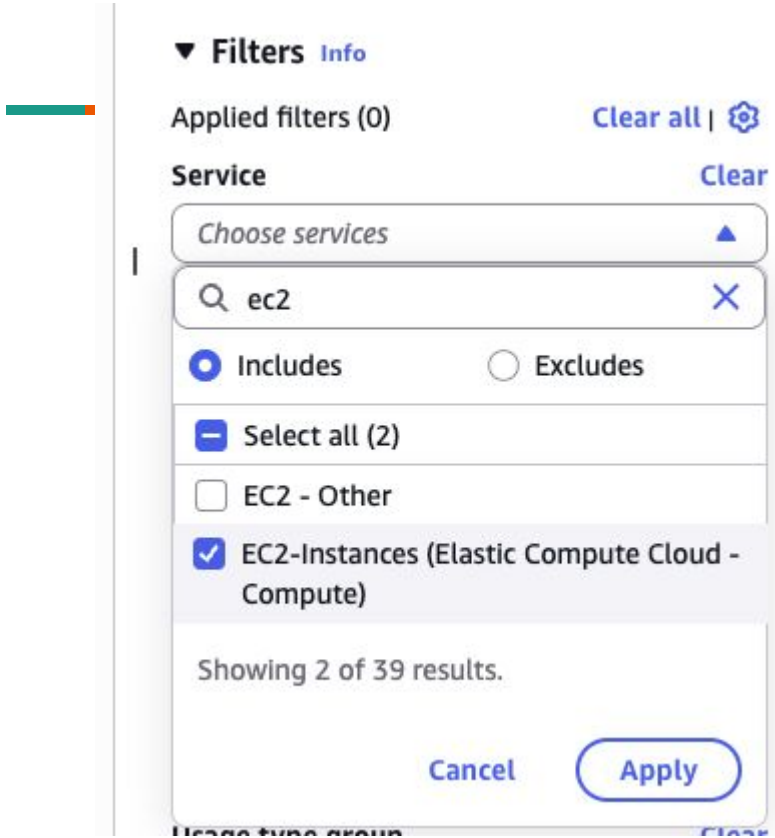
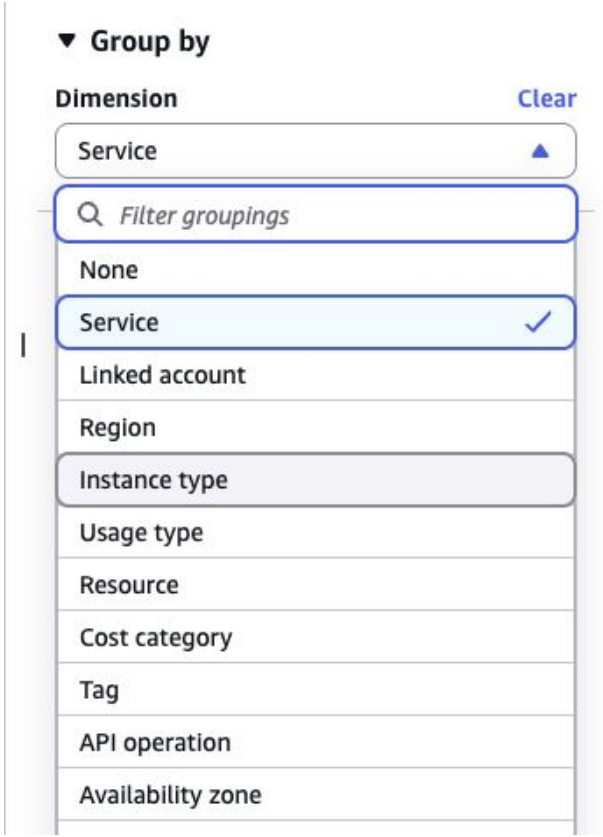
EC2-Instances
OpenSearch Service
EC2-Other
Relational Database Service
VPC
S3
Elastic Container Service for Kubernetes
Config
GuardDuty
Others


Filters Info
Applied filters (0)
Clear all
Service
Clear
Choose services
ec2
Includes
Excludes
Select all (2)
EC2 - Other
EC2-Instances (Elastic Compute Cloud - Compute)
Showing 2 of 39 results.
Cancel
Apply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Linked account
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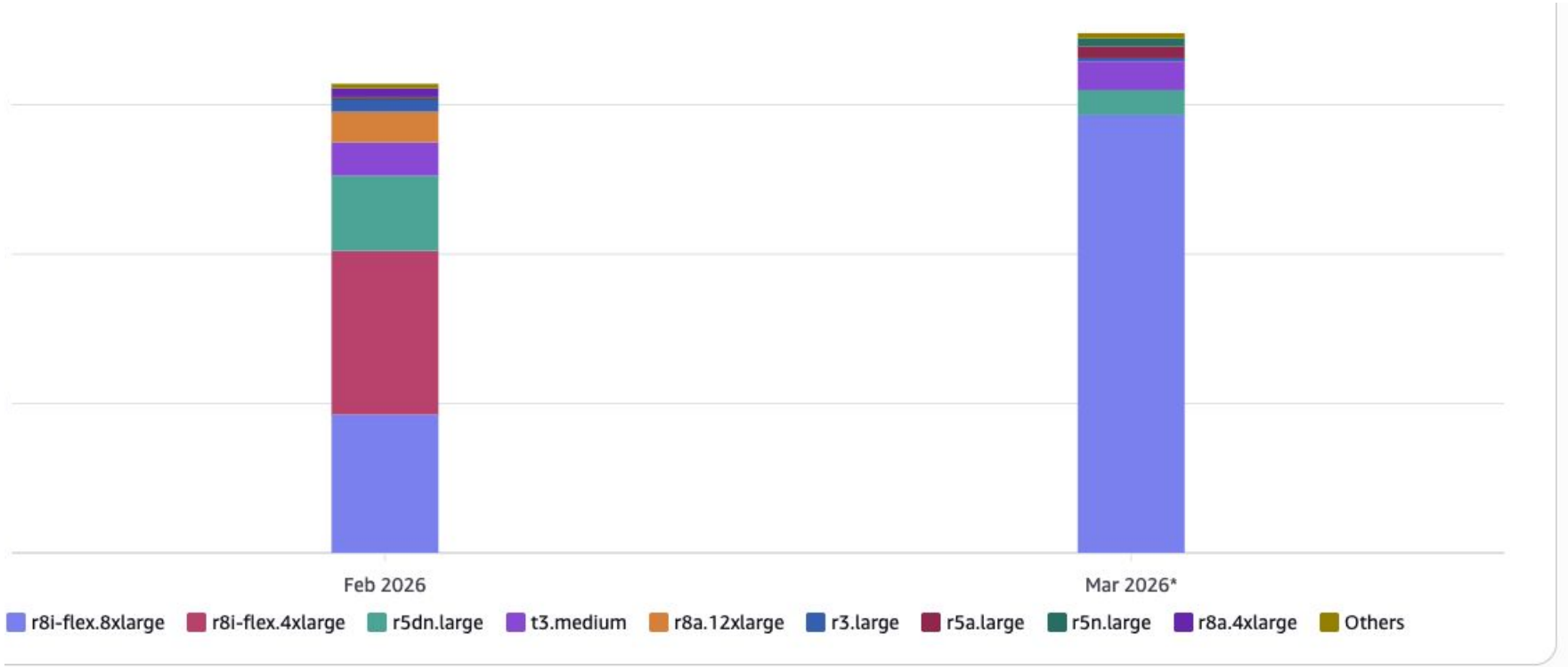
API operation
Availability zone


Feb 2026
Mar 2026*
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r8i-flex.4xlarge
r5dn.large
t3.medium
r8a.12xlarge
r3.large
r5a.large
r5n.large
r8a.4xlarge
Others

Follow along?

- We will be making changes/updates to the cluster-level-resources charts in gen3-helm

Future:

- Graviton Instances with EKS 1.35 update
- Potentially dropping VPC endpoints

Q & A